

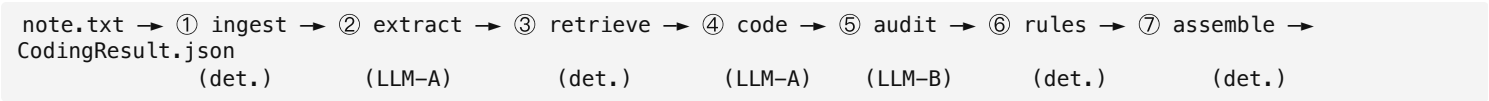
medcoder — Auditable Medical-Coding Pipeline

AppliedAI · Opus AI Engineer · Exercise 2
Design Document (1–2 pp distillation)
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1 · Architecture & data flow

Design thesis. The LLM is a *constrained reasoning component inside a deterministic, observable pipeline* — not the pipeline itself. We make a stochastic model auditable by **pre-constraining it with retrieval** (it can only choose real codes), **independently verifying it with an auditor agent**, and **post-constraining it with a deterministic rule engine** grounded in the ICD-10-CM Official Guidelines.

The pipeline has seven stages, each idempotent and independently retryable:



- Ingest** — normalize, detect encounter type (inpatient vs outpatient matters: it governs whether “probable / suspected” diagnoses are codable per ICD-10-CM Guideline IV.H), segment SOAP sections, window long multi-page notes with overlap, preserve **global character offsets**.
- Extract (LLM-A)**. Returns `ExtractedFact[]` — verbatim span, normalized clinical term, **assertion status** (present / absent / possible / hypothetical / family / historical), kind (dx / px / symptom). A deterministic `NegEx/ConText`-style backstop overrules clear LLM polarity slips (“denies chest pain” must never be coded as active chest pain).
- Retrieve** — hybrid search per fact (see §2).
- Code (LLM-A)**. Picks 1–N codes **only from the retrieved whitelist**, with verbalised confidence and a rationale that quotes the evidence span.
- Audit (LLM-B independent)**. Re-reads each (evidence, code) pair against the cited evidence + code description and returns agree/disagree + note. *Selective + batched*: only procedures + low-confidence diagnoses go through the heavy auditor.
- Rules (deterministic)**. Excludes1 conflicts, unspecified-when-specific, missing 7th character on injury/pregnancy chapters, evidence anchoring, procedure-without-supporting-diagnosis, format validity. Emits typed `Warning[]` (missing_information / ambiguity / conflict).
- Assemble (deterministic)**. Blends and tiers the confidence (§3), builds the Pydantic-validated `CodingResult` and a `RunMetadata` envelope with `trace_id`, model snapshots, prompt version, config hash, and per-stage metrics.

2 · Code retrieval / filtering strategy

The single biggest failure mode for full-vocabulary ICD-10 coding is the LLM **hallucinating non-existent codes** — across GPT-4/Gemini/Llama models, up to 35% of generated ICD-10 codes are non-billable or invalid (NEJM AI, 2024). We eliminate this structurally: the LLM never free-generates a code.

For each extracted fact (using its `normalized_term` as the query):

- Dense search** with `sentence-transformers/all-MiniLM-L6-v2` over a FAISS inner-product index (catches paraphrases — “DM type 2” finds “type 2 diabetes mellitus”). Returns top-N=50.
- Lexical BM25** over the same descriptions (catches exact clinical phrasing — “essential hypertension” matters word-for-word). Returns top-N=50.
- Reciprocal Rank Fusion** (Cormack et al. 2009): each retriever contributes $1 / (k + \text{rank})$; fused scores are summed. RRF needs **no score calibration** between heterogeneous scorers — the empirically dominant hybrid fusion choice. Top-K=15 candidates become the **whitelist**.

That whitelist is a **hard constraint**, not soft context. The coder agent’s response is re-validated against the candidate set; any out-of-list code is dropped and logged. This makes hallucinated codes *structurally impossible*.

Data: we bundle the **real CDC ICD-10-CM FY2027 catalog** (~74,879 codes, US public domain) — using a synthetic ICD-10 catalog would dodge the large-code-space challenge the exercise tests. CPT is **AMA-copyrighted with no free tier**, so we ship a clearly-marked **synthetic CPT-shaped catalog** and make licensed real CPT a one-config-line drop-in.

Production swap (documented): the same `Retriever` interface backs equally well by **Postgres pgvector + tsvector + pg_trgm** — one datastore for semantic + full-text + fuzzy in one operational footprint.

3 · LLM usage & prompting approach

Three role-specialized agents (mirroring the real-world coder → QA-auditor workflow), all behind a single **LiteLLM** gateway so providers swap by env var:

Agent	Human analog	Model (default)	Constraint
Extraction	clinical scribe	openai/gpt-4o-2024-08-06	reason-then-format; assertion regex
Coder	medical coder	openai/gpt-4o-2024-08-06	may only choose from whitelist
Auditor	QA auditor	anthropic/claude-3-5-sonnet-20241022	different model family by default

The **auditor defaulting to a different model family** is deliberate — research shows heterogeneous verifiers cut correlated errors and self-preference bias (arXiv 2410.21819; A-HMAD 2025). Same-model fallback is supported but flagged as weaker. **Selective verification**

(only procedures + low-confidence diagnoses) and **batched calls** (one extract / one code / one audit per note) keep the cost discipline reasonable.

Prompting choices:

- *Reason-then-format*. Strict JSON-only constraints degrade reasoning quality (arXiv 2408.02442). Prompts instruct the model to think internally first, then emit a single JSON object validated against a Pydantic schema. On validation failure we **repair-retry** once with the error appended to the conversation — almost always sufficient.
- *Versioned prompt files* (`prompts/extraction_p1.txt`, etc.). Prompt versions are part of the `config_hash` so a prompt change is visible in the audit log.
- *temperature=0 + pinned dated model snapshots* (e.g. `openai/gpt-4o-2024-08-06`) — reproducibility-first, traded off against the marginal accuracy of self-consistency sampling. Self-consistency is available as an optional confidence signal but off by default.

Confidence we surface ≠ raw LLM confidence. Verbalised LLM confidence is systematically overconfident (Xiong et al. 2023). We blend three signals — retrieval rank (rank-aware, not raw RRF score), coder confidence (discounted), and an auditor adjustment (+0.15 for agree, -0.30 for disagree) — and bin into ● / ● / ● tiers using gold-tuned thresholds. (Formal isotonic/Platt calibration is wired as an extension — needs a larger labelled set than the demo gold supports.)

4 · Key decisions & trade-offs

Decision	Why	Trade-off accepted
Retrieve-then-constrain (whitelist, not free generation)	Eliminates hallucinated codes; turns a 75k-way generation problem into a k-way selection (6% → ~100% on a comparable task — arXiv 2407.12849)	Bounded by retriever recall@k — the upstream recall ceiling on retrievable codes (not computed in the eval)
Coder + independent Auditor (multi-LLM)	Best precision/recall on MIMIC-IV (MDPI Informatics 2026); heterogeneous verifiers cut correlated errors (A-HMAD 2025)	Extra LLM calls — mitigated by selective+batched verification
Hybrid retrieval + RRF	Exact terms <i>and</i> paraphrases; no score-scale tuning	Two indexes to build; ~1 minute on 75k codes
Bounded 3-agent decomposition (no swarm)	MAST (NeurIPS 2025) shows speculative swarms add latency and “silent gray errors” without reliable accuracy gains	Foregoes ensemble-style accuracy gains for predictability
Real ICD-10 + synthetic CPT	Solves the brief’s large-code-space challenge; legal under AMA licensing	CPT demo runs on synthetic codes
LiteLLM gateway	Multi-provider in one call; built-in <code>mock_response</code> for keyless tests	Thin dependency
temp=0 + pinned snapshots over sampling	Reproducibility / auditability is an explicit requirement	Foregoes ~1–3 pt accuracy from self-consistency (offered as optional)
Deterministic rule engine for hard checks	Auditable, exact, no LLM cost	Full guideline coverage needs the ICD-10 tabular XML + CMS NCCI
CLI-only core (FastAPI/UI as extensions)	Matches the offline brief; no over-build	No reviewer UI — but the JSON contract supports override in-place
Blended-then-tiered confidence	Raw LLM confidence is systematically overconfident	Calibration is gold-tuned, not formally calibrated (extension)

5 · Limitations & extensions

Limitations (honest). *Assistive, not autonomous* — SOTA full-vocabulary ICD-10 tops out around micro-F1 ≈ 0.54 (RAG-Coding, 2026); a human reviewer is required by design and the payload (evidence spans, tiered confidence, mutable reviewer fields) is shaped to make that review fast. *CPT is synthetic* — real-CPT accuracy must be re-validated on a licensed catalog (architecture is drop-in). *General-purpose embedder* (`all-MiniLM-L6-v2`) is a demo compromise; production choice is biomedical (SapBERT / PubMedBERT). *Rule engine is a curated subset* (format, billable, Excludes1 short list, 7th-character chapters, dx↔px linkage, evidence anchoring); full guideline coverage needs the ICD-10 tabular XML + CMS NCCI refresh pipeline. *Confidence calibration* is gold-tuned thresholds, not formal Platt / isotonic — needs a larger labelled set. *Reproducibility* is engineered (temp=0, pinned dated snapshots, versioned prompts, full audit log) but not bit-for-bit guaranteed across provider model updates — exactly why we pin and log everything. *Evaluation* is illustrative on a small authored gold set; the metric methodology (`scripts/evaluate.py` : micro P/R/F1 for ICD and CPT, exact-match ratio, ICD-10-hierarchical micro-F1) is correct; only the sample size is small.

Extensions (designed for, not built). Postgres hybrid retrieval (`pgvector` + `tsvector` + `pg_trgm`); biomedical embeddings + SNOMED→ICD crosswalk; full tabular-rule + NCCI engine; FastAPI + reviewer UI; self-consistency confidence; AHIMA/ACDIS-compliant provider-query drafting; licensed real CPT (one config line); LLM-observability tooling (Langfuse / OpenTelemetry export via LiteLLM callbacks — opt-in env flag; Grafana + Prometheus dashboards in production). Pipeline stages are idempotent and map cleanly to **Airflow tasks** or **Celery jobs** for production orchestration.

References (load-bearing). Soroush et al. *LLMs Are Poor Medical Coders* (NEJM AI 2024); *Code Like Humans* (EMNLP 2025, 2509.05378); MDPI Informatics 2026 (coder+auditor); 2407.12849 (retrieve-then-rerank); MAST (NeurIPS 2025); Huang et al. *Can’t Self-Correct Reasoning Yet* (ICLR 2024, 2310.01798); Chain-of-Verification (2309.11495); Cormack et al. 2009 (RRF); 2408.02442 (JSON ≠ free-form); Xiong et al. 2306.13063 + Tian et al. 2305.14975 (confidence); ICD-10-CM Official Guidelines FY2026 (CMS/CDC); AMA CPT licensing FAQ.